

Interplay of social and actuarial issues for social security in the Caribbean

Population dynamics, aging trajectories, fertility shifts

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Linguistic & Cultural Sub-Regions

Linguistic Zone	Primary Countries	Population Share	Dominant Languages
Spanish Caribbean	Cuba, Dominican Rep, Puerto Rico	57.5%	Spanish
French / Creole	Haiti, Guadeloupe, Martinique	28.0%	French, Haitian Creole
Anglo Caribbean	Jamaica, Trinidad, Barbados, Bahamas	13.5%	English, English Creoles
Dutch Caribbean	Aruba, Curaçao, Sint Maarten	1.0%	Dutch, Papiamentu

Key Demographic Shifts in Caribbean



Sub-Replacement Fertility

TFR dropped below 2.1 in major islands including Cuba, Puerto Rico, and Jamaica.



Accelerated Aging

Elderly population ratio rising rapidly in Cuba and Puerto Rico due to low births.



Persistent brain drain to North America balancing local unemployment pressure.



Sub-Replacement Fertility

TFR dropped to 1.4–1.6 across most OECS islands, well below the 2.1 replacement benchmark.



Accelerated Aging

Median age rising rapidly past 36 years in Barbados, St. Kitts, and Saint Lucia.



Non-Marital Family Structure

Visiting unions and common-law partnerships dominate, with low formal crude marriage rates.

Total Fertility Rate (TFR) Comparison

Country	1990 TFR	2010 TFR	2026 TFR	Status vs Replacement (2.1)
Barbados	1.8	1.5	1.3	Critically Low
Saint Lucia	3.1	1.8	1.4	Below Replacement
Antigua & Barbuda	2.2	1.7	1.4	Below Replacement
St. Kitts & Nevis	2.8	1.7	1.5	Below Replacement
Grenada	3.4	2.0	1.5	Below Replacement
Dominica	3.0	1.9	1.6	Below Replacement
St. Vincent & Grenadines	3.2	2.1	1.7	Moderate Low
Belize	4.5	2.6	2.0	Near Replacement

Total Fertility Rate (TFR) Comparison

Replacement level fertility is 2.1 births per woman. Most Caribbean nations sit below this line.

Country	TFR (1990)	TFR (2010)	TFR (2024/2025)	Status vs Replacement (2.1)
Haiti	5.2	3.5	2.7	Above Replacement
Dominican Republic	3.6	2.6	2.2	Near Replacement
Jamaica	3.1	2.2	1.8	Below Replacement
Trinidad & Tobago	2.5	1.8	1.6	Below Replacement
Cuba	1.8	1.5	1.4	Critically Low
Puerto Rico	2.2	1.6	0.9	Ultra Low

Larger Caribbean Populations by Country

Country / Territory	Population (2024/2025)	Share of Region	Annual Growth Rate	Urbanization Rate
Haiti	11.72 Million	26.3%	+1.12%	59.8%
Dominican Republic	11.33 Million	25.5%	+0.92%	83.2%
Cuba	11.02 Million	24.8%	-0.35%	77.4%
Puerto Rico (US)	3.22 Million	7.2%	-0.48%	93.6%
Jamaica	2.83 Million	6.4%	+0.15%	56.7%
Trinidad & Tobago	1.53 Million	3.4%	+0.28%	53.2%

Median Age (2026 Estimates)

Eastern Caribbean islands show advanced demographic aging compared to Central American Belize.



Marriage Rates & Union Formations



Low Marriage Rates

Crude marriage rates across the Eastern Caribbean range between **2.5 to 4.2 per 1,000 residents**.

Formal marriage is increasingly delayed until mature adulthood or bypassed in favor of informal unions.



Prevalent Union Types

Visiting Unions: Partners maintain separate residences while bearing children.

Common-Law Unions: Co-habiting non-marital partnerships.

Matrifocal Units: High proportion of female-headed households.

Marriage & Family Structure

Country	Crude Marriage Rate	Out-of-Wedlock Births (%)	Avg Household Size	Dominant Union Structure
Saint Lucia	3.2 / 1,000	82%	2.9 persons	Visiting / Single Mother
St. Kitts & Nevis	3.5 / 1,000	78%	3.0 persons	Visiting / Extended
Grenada	2.9 / 1,000	81%	3.2 persons	Matrifocal / Common-Law
Antigua & Barbuda	4.1 / 1,000	74%	3.1 persons	Single Parent / Extended
Dominica	3.0 / 1,000	79%	3.1 persons	Common-Law / Visiting
Belize	4.8 / 1,000	62%	3.8 persons	Nuclear / Common-Law

Out-of marriage: USA: 40%, Chile: 78%, México: 73%, Colombia: 87%, Japan 2.4%. Many in common law unions.
Still, many births out of marriage and out of unión: Chile: 25%, Colombia: 21%

Elderly Dependency & Healthcare Strain

Countries with highest elderly dependence: Puerto Rico, Martinique & Guadeloupe, Curaca & Aruba, Barbados, Cuba

Countris with lowest elderly dependence: Honduras, Haiti, Guatemala, Belize, Bolivia

65+ POPULATION SHARE (2030 PROJ.)

18.5%

Average across Barbados, St. Kitts, Antigua, and Saint Lucia.



Pension Solvency Pressure

Fewer working-age contributors funding national insurance schemes.



NCD Care Burden

High prevalence of hypertension and diabetes requiring long-term care.

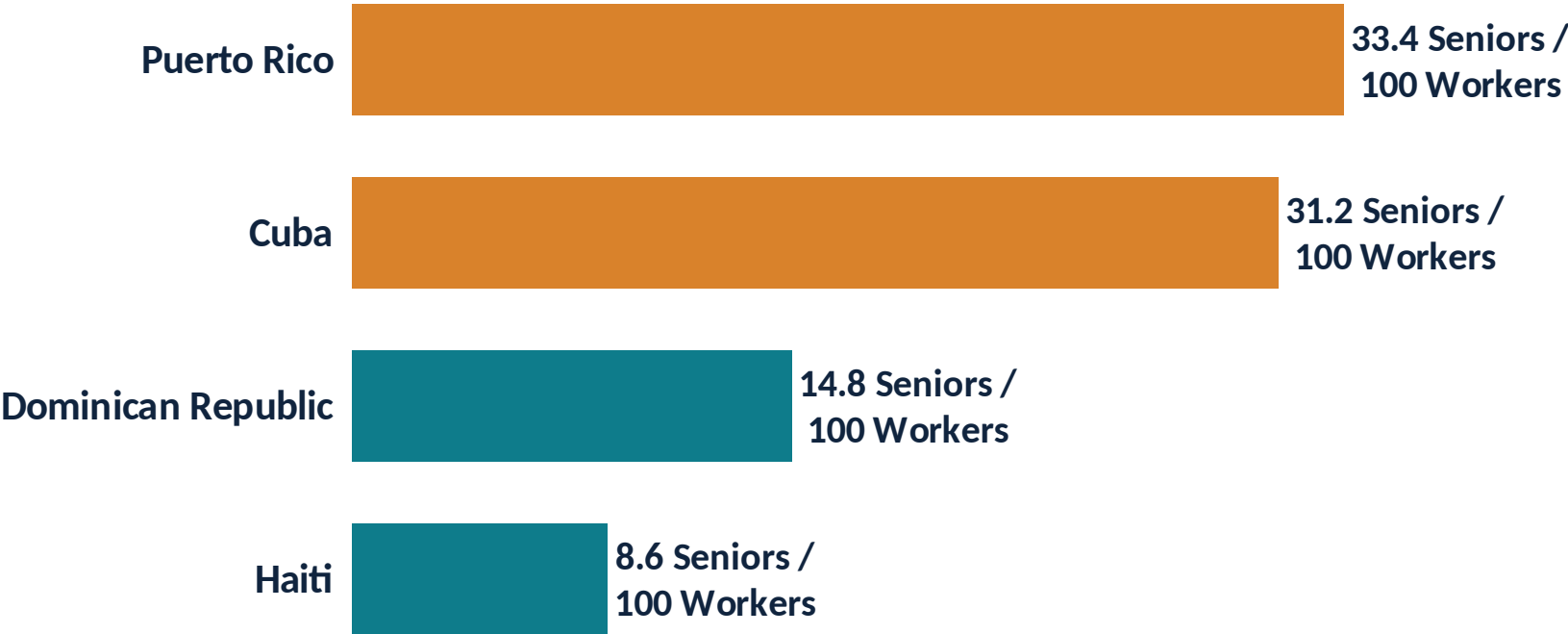


Sandwich Generation

Single mothers balancing elderly parent care and child support.

Old-Age Dependency Ratio Pressure

Dependents per 100 working-age adults (15-64). Rising sharply in older islands.



Drivers of Declining Marriage & Lower Fertility

1

Female Autonomy

Higher tertiary enrollment among women leading to delayed childbearing.

2

Out-Migration

Youth brain drain disrupts local partner pools and union stability.

3

Cost of Living

Housing expenses lead couples to postpone formal marriage.

4

Cultural Norms

Historical acceptance of visiting unions as valid child-rearing units.

Migration Dynamics & Remittance Patterns



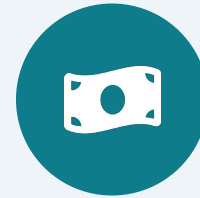
Labor Outflow

High emigration rates toward US, Canada, and Europe.



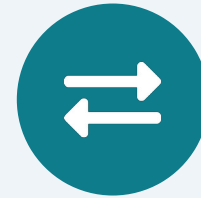
Brain Drain

Over 60% of tertiary-educated citizens move abroad.



Remittance Inflow

Remittances account for over 20% of GDP in Haiti and Jamaica.



Intra-Regional

Haitian movement into Dominican Republic and Bahamas.

Cuba: Demographic Crisis & Aging Population



POPULATION

11.02 M

Annual shrinking rate of -0.35%

FERTILITY

1.4 TFR

OVER 65 YRS

21.6%

Puerto Rico: Rapid Population Decline



-14.2%

Population drop since 2010 due to economic exodus to US mainland and ultra-low births (0.9 TFR).

Jamaica: Stagnation & Youth Outflow



Informal economy

Large informal economy, low coverage, large number of elderly with no pension



Non-contributory pensions

Relatively recent programs to support elderly with no pension



Zero Growth

Population stable at 2.83M due to high outward migration balancing natural increase.



Skilled Labor Deficit

Teachers and healthcare workers actively recruited by US and UK institutions.



Urban Shift

56.7% urbanized with heavy focus in Kingston Metropolitan Area.

Trinidad & Tobago: Demographics



Fast aging (39 year median age)



Large immigration from Venezuela in recent years.



Fertility rate: 1.6 children



Significant fiscal support



The labor force is contracting



Significant informal economy



High levels of out-migration by college-educated



Most retirees receive the minimum NIS pension

Eastern Caribbean States (OECS Demographics)

Barbados

282k pop.

Advanced aging, high life expectancy (77.8 yrs).

Saint Lucia

180k pop.

Moderate growth, tourism-driven urban drift.

Grenada

125k pop.

High diaspora population in North America.

Antigua & Barbuda

100k pop.

Immigration destination for regional workers.

Policy Interventions for Small States



Childcare Support

Subsidize early childhood centers to support working single mothers.



Pension Adjustments

Gradually align retirement ages with expanding life expectancy.



Legal Equality

Protect property and inheritance rights in common-law partnerships.



Youth Retention

Create remote tech jobs to curtail young adult brain drain.

Epidemiological Transition Indicators



NCD Burden

Non-communicable diseases (diabetes, cardiovascular) cause over 75% of deaths in Anglo Caribbean.



Diabetes Prevalence

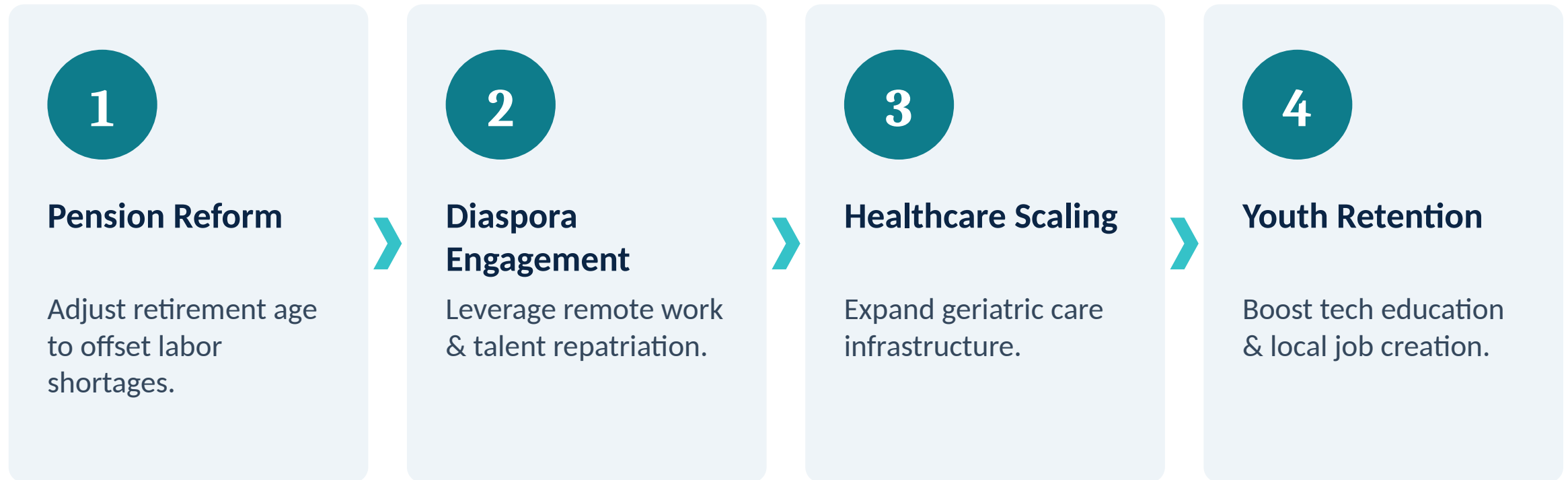
Caribbean has among the world's highest rates (~12-15% adult prevalence).



Infant Mortality Gap

Rangers from 5.1/1000 in Cuba to 48.0/1000 in Haiti.

Demographic Policy Action Framework



Summary: The Demographic Horizon



1. Persistent Low Fertility

OECS states have firmly entered low fertility regimes, accelerating population decline risks.



2. Non-Traditional Family Norms

Formal marriage is a secondary family structure; visiting and common-law unions prevail.



3. Aging Care Deficit

Small island budgets require structural healthcare and pension adaptation before 2035.

Challenges, social and financial

Economic and actuarial views

It is not possible to predict the evolution of key variables in the long term

Jeremy Greenwood, Nezih Guner and Guillaume Vandenbroucke. Family Economics Writ Large. FEDERAL RESERVE BANK OF ST. LOUIS Working Paper Series Working Paper 2016-026B <https://doi.org/10.20955/wp.2016.026>. January 2017

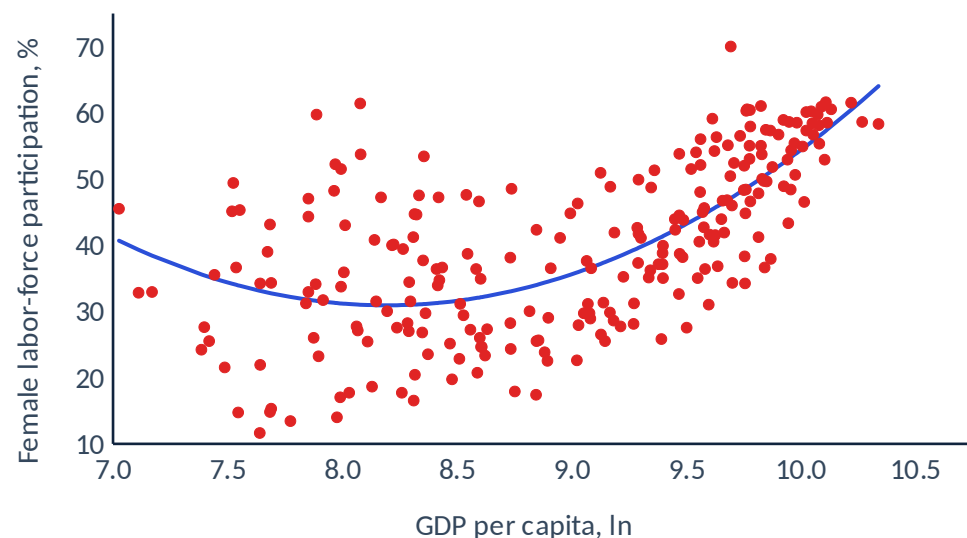


Figure 4. Economic Development and Female Labor-Force Participation for a Panel of Countries, 1890-2005

With development, there is more work by women, but there is a great dispersion over time and between countries.

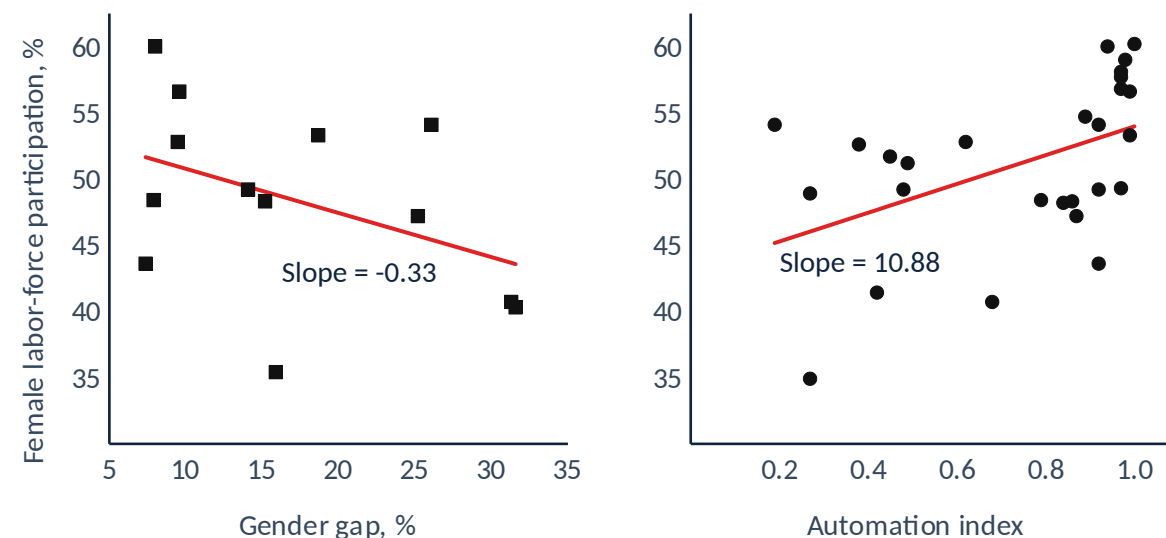


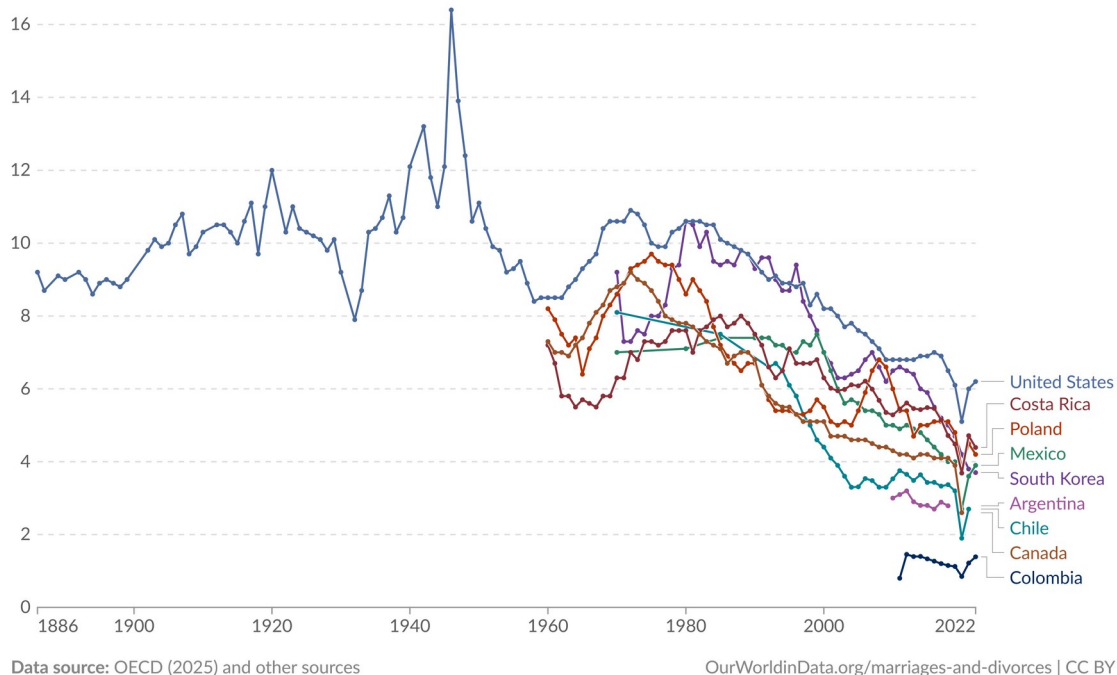
Figure 6. Gender Gap and Household Technologies versus Female Labor-Force Participation for a Cross Section of Countries

Automation has historically favored women's work (cheaper household appliances). This is not easy to predict, and we are currently living in an era of high automation due to digitalization and AI.

Another example of a variable that is very difficult to predict: marriage

Marriage rates

Number of marriages during a given year per 1,000 people.



OECD (2025), OECD (2024), US Census Bureau (1949), Centers for Disease Control and Prevention (2020) – with minor processing by Our World in Data

A relatively recent example of the difficulty of predicting behavior, one of great relevance: the fall in marriage rates and the growth of births out of wedlock.

Initially, it seemed that it was just an elevation of the age of marriage.

Now, it seems that there is a significant increase in women who never get married, and those who have children in free unions.

The great family events of the last decades



Huge growth in women's labor force participation rates.



Great decrease in marriage and increase in divorce.



Increase in positive pairing in marriages (the most educated marry in increasing proportions with the most educated).



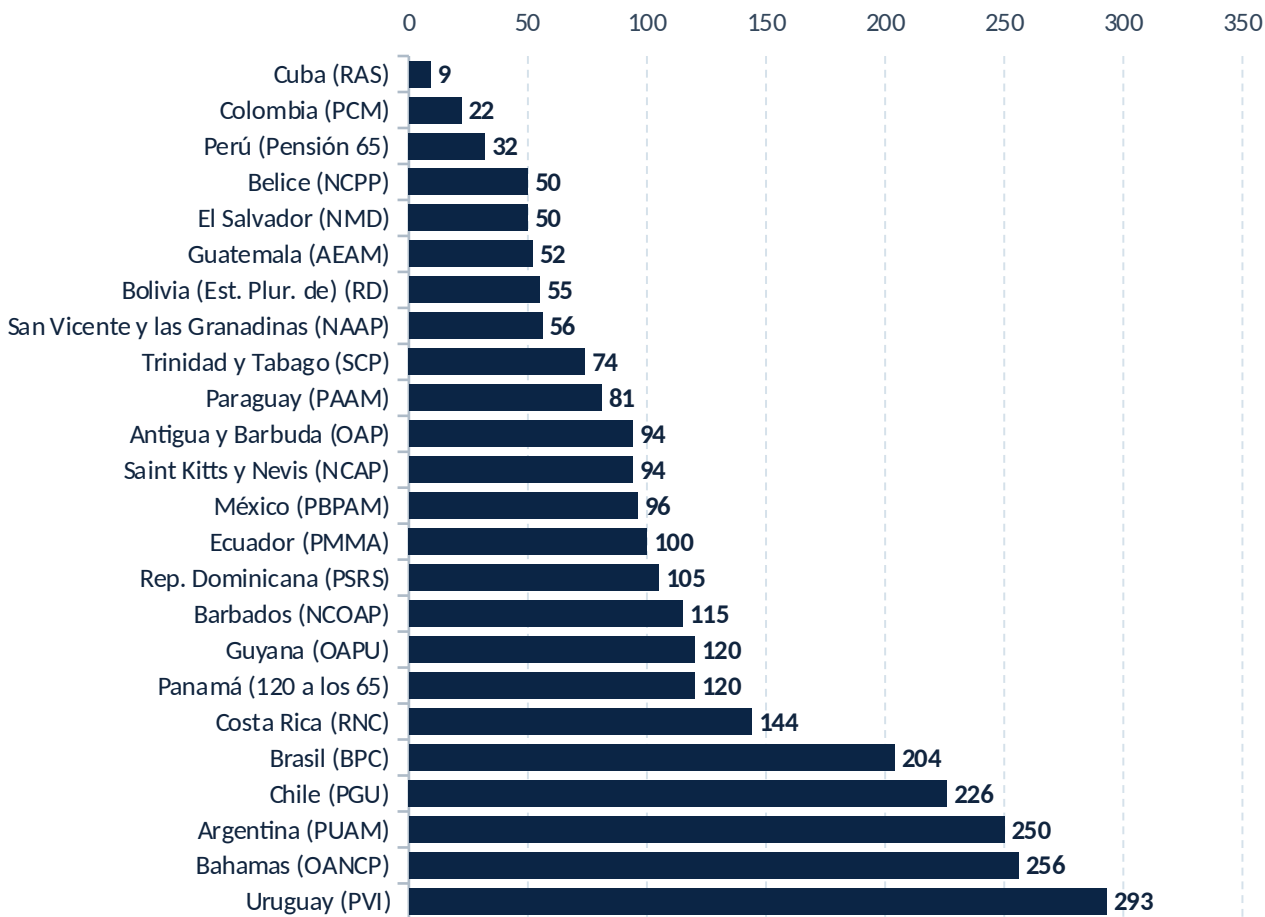
Growing up children in poverty with a parent (usually a single mother)



Explosion and fall in fertility

Countries found non-contributory pensions as the only way to face the huge political pressure from aging populations with a large share of uninsured

A. Prestaciones de vejez^b



B. Prestaciones de discapacidad

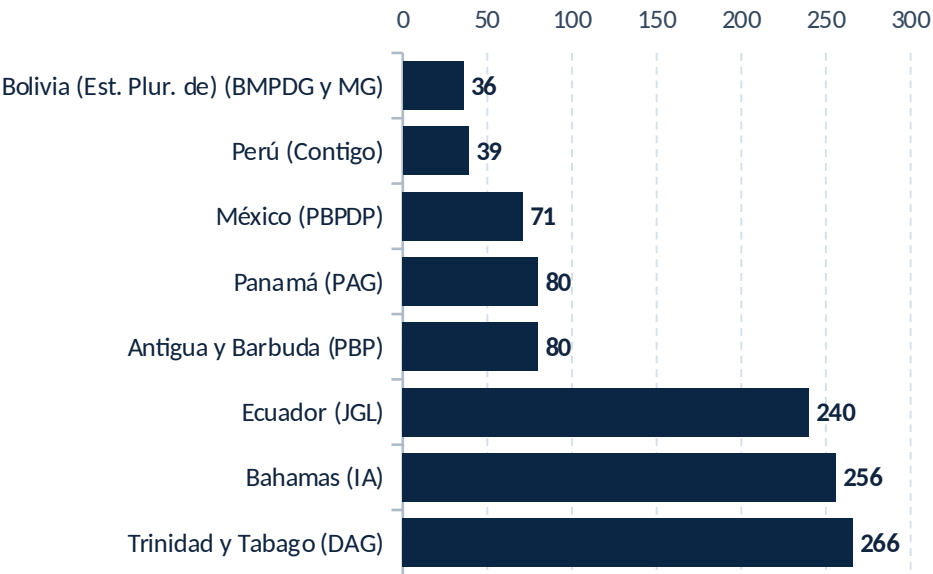


Gráfico II.2
América Latina y el Caribe (24 países): valor mensual de las pensiones no contributivas, 2021 o último año con información disponible^a
(En dólares)

Generalized historical problems

Historically, pension systems faced challenges due to

- difficulties in financing the payment of benefits,
- for not adequately dealing with deep episodes of unemployment,
- for paying pensions so low that they generated dissatisfaction among the population,
- for operating disordered concepts of invalidity,
- for paying such high pensions to some that they discouraged work and generated dissatisfaction from those who did not receive those benefits,
- for not recognizing structural changes in families (divorce trends, for example),
- for the use of resources for purposes of the state other than paying benefits,
- for systematically failing to adapt to population aging,
- for not responding adequately to changes in the financial environment, such as inflation or variations in interest rates,
- for not having adequate schemes for workers in high-turnover occupations or non-subordinate working conditions.

Hypothesis



The motivation for reforms does not come from innovations in systems design, but from the fact that social dynamics exceed the capacity of the original design to cope with social changes.

It is not possible to predict how women's fertility rates and work, longevity and the need for long-term care, financial costs, tax legislation, women's labor force participation, and other factors that affect the sustainability of systems will evolve.

Social security in the countries of the Americas was founded on a certain expectation that labor markets would evolve to the predominance of salaried labor and that families would be predominantly nuclear. The reality by the 1970s was already one of a significant informal economy and changes in family organization, with high levels of divorce, out-of-wedlock births, and a significant fraction of work in non-salaried formats, largely informal



Shall we rely on actuarial or economic approaches to study the challenges and propose policy?

Actuarial methods	Economic methods
Goal: measurement of financial flows, actuarial liabilities and risks, including future evolution under scenarios	Goal: to identify causality of events and prediction
High internal consistency of models for complete systems	Causal Approach: Using "Reduced Forms" to Capture Relationships Prediction approach: short-term models
Consistent demographic and financial models	There is generally no pretension of modeling systems integrally
Ability to measure accurately, under assumptions. High reliability when there is control capability (e.g. in a private insurance plan)	In causal approaches, the main thing is to identify the validity of the assumptions. In prediction approaches, the important thing is the error of the prediction, regardless of the assumptions.
Actuaries usually work for governments, insurers, other financial institutions, social security agencies, and consulting firms, where they financially manage their health, casualty, life, and property-related insurance policies	Economists usually work for the public and financial sectors, consulting, and increasingly for all types of companies that use data for management, sales, and other processes. Most are not specialists in econometrics and tend to specialize in sectors (e.g., banking, agriculture, ecology, industry, antitrust, labor, health, etc.).

CONCLUSION & DISCUSSION

Interplay of social and actuarial issues for social security in the Caribbean

Balancing social protection, labor retention, and family welfare policies across the Caribbean.

Questions & Discussion